Chapter 4: Prompt Engineering

Learning Objectives

By the end of this chapter, you will:

- Understand the fundamentals of Prompt Engineering and its role in AI applications.
- Learn various prompting techniques used in enterprise environments.
- Develop frameworks for managing prompts, evaluations, and governance as an AI TPM.

Beginner Explanation

Prompt Engineering is the process of designing instructions that help an AI model generate useful, accurate, and consistent responses.

Think of prompts as requirements documents for AI.

If you ask:

> "Tell me about Azure."

You will receive a broad answer.

If you ask:

> "Act as an Azure Solutions Architect and explain Azure OpenAI to a Technical Program Manager in less than 100 words."

You receive a much more targeted and useful response.

Why Prompts Matter

Poor Prompt	Better Prompt
Write an email.	Write a professional email to a VP requesting budget approval for an AI project.
Explain AI.	Explain Generative AI to a Delivery Manager using practical examples.
Create a report.	Generate a weekly TPM status report with risks, blockers, and next steps.

Prompt Formula

A good prompt usually includes:

1. Role
2. Context
3. Task

4. Constraints
5. Output Format

Example

```
Role:
You are a Senior Technical Program Manager.

Context:
An enterprise is migrating from Teradata to Azure.

Task:
Create a weekly status report.

Constraints:
Less than 200 words.

Output:
Table format.
```

Intermediate Explanation

Prompt Engineering has become a critical capability for enterprises because it directly impacts:

- Accuracy
- Cost
- User experience
- Response consistency
- Productivity

Most organizations maintain centralized prompt libraries.

Common Prompting Techniques

Technique	Purpose
Zero-Shot	No examples provided
One-Shot	Single example provided
Few-Shot	Multiple examples provided
Chain of Thought	Step-by-step reasoning
Self-Consistency	Multiple reasoning paths
Structured Output	JSON, XML, Tables
Role Prompting	Assigning personas

Enterprise Prompt Lifecycle

1. Design
2. Test
3. Review
4. Deploy
5. Monitor
6. Optimize

AI TPMs are increasingly responsible for:

- Prompt governance
- Prompt versioning
- Evaluation frameworks
- Change management
- KPI reporting

Why This Matters for an AI TPM

Responsibility	Impact
Delivery	Ensures AI applications produce reliable outputs.
Risk	Reduces hallucinations and inconsistent responses.
Stakeholder Management	Aligns business expectations with AI capabilities.
Budget	Optimizes token consumption through efficient prompts.

Enterprise Example

Business Problem

A global consulting organization wanted to automate weekly project status reports across 200+ active programs.

Solution

Develop a Prompt Engineering framework that generates:

- Weekly reports
- RAID summaries
- Executive updates
- Steering committee decks

Architecture

- Azure OpenAI
- Azure AI Foundry

- Azure Functions
- Azure DevOps
- SharePoint

Outcome

Metric	Result
Time Saved	15 Hours per TPM per Week
Report Accuracy	93%
User Adoption	89%
Annual Productivity Gain	~12,000 Hours

Lessons Learned

- Prompt quality directly impacts business outcomes.
- Version control is necessary.
- Structured outputs improve reliability.
- Continuous evaluations are essential.

Azure Implementation

Services Used

Service	Purpose
Azure OpenAI	Prompt execution
Azure AI Search	Enterprise context retrieval
Azure Functions	Workflow orchestration
Azure AI Foundry	Prompt management and evaluations

Implementation Workflow

Step 1: Create Prompt Templates

Examples:

- Executive Summary Prompt
- Meeting Notes Prompt
- Risk Assessment Prompt
- Status Report Prompt

Step 2: Build Prompt Library

Prompt Name	Owner	Version
Weekly Report	TPM Office	v1.2
Risk Summary	PMO	v1.0
Executive Update	Leadership Team	v2.1

Step 3: Define Evaluation Criteria

Track:

- Accuracy
- Completeness
- Latency
- Cost
- User Satisfaction

Step 4: Implement Governance

Establish:

- Approval processes
- Prompt reviews
- Version control
- Audit logs

Mermaid Diagram

```
graph TD

A[Business Requirement]

A --> B[Prompt Template]

B --> C[Azure OpenAI]

C --> D[Azure AI Search]

D --> E[Generated Output]

E --> F[Application]

F --> G[End Users]
```

TPM Checklist

- [ ] Define prompt standards.
- [ ] Create a prompt library.
- [ ] Establish version control.
- [ ] Define evaluation metrics.
- [ ] Conduct user testing.
- [ ] Implement governance controls.
- [ ] Monitor token consumption.
- [ ] Create rollback procedures.
- [ ] Document lessons learned.
- [ ] Train business users.

Templates

Prompt Review Template

Item	Status
Business Objective Defined	Yes/No
Context Included	Yes/No
Constraints Added	Yes/No
Output Format Defined	Yes/No
Evaluation Completed	Yes/No

Prompt Library Register

Prompt	Owner	Version	Status
Weekly Status	PMO	v1.0	Active
Risk Assessment	TPM Team	v1.1	Active
Executive Summary	Leadership	v2.0	Active

Evaluation Scorecard

Metric	Target
Accuracy	>95%
Latency	<3 Seconds
Cost	<$0.05 per Request

Metric	Target
Satisfaction	>90%
Reliability	>99%

Common Mistakes

1. Writing vague prompts.
2. Ignoring output formatting requirements.
3. Failing to version prompts.
4. Not testing prompts across multiple scenarios.
5. Assuming prompts remain effective indefinitely.
6. Ignoring token costs.
7. Skipping governance and approvals.

Key Takeaways

- Prompt Engineering is one of the most important skills for AI TPMs.
- Better prompts lead to better business outcomes.
- Enterprises should maintain centralized prompt libraries and governance processes.
- Structured prompts improve consistency, reliability, and cost efficiency.
- Azure AI Foundry provides tools for prompt management and evaluations.
- Prompt Engineering should be treated as an enterprise capability rather than an individual skill.

Footer

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charan.ai | Page 17